

Rationale of 5G Requirements

1. Definition

- Rationale of 5G means the **reasons why 5G technology is required** and what requirements it must satisfy.
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2. Why do we need 5G? (Core Idea)

 Because:

- 4G is **not enough** for future demands
 - Need **high speed + low delay + more devices**
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4. Key Requirements of 5G (Explain These Points)

1. High Data Rate

- Up to **10 Gbps**
 - Needed for:
 - HD video
 - VR/AR
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2. Low Latency

- Delay as low as **1 ms**
- Required for:
 - Online gaming
 - Remote surgery

◆ 3. Massive Connectivity

- Supports **1 million devices/km²**
- Needed for:
 - IoT (smart homes, cities)

◆ 4. High Reliability

- Very low failure rate
- Important for:
 - Healthcare
 - Autonomous vehicles

◆ 5. Energy Efficiency

- Low power consumption
- Needed for battery devices

◆ 6. High Spectral Efficiency

- Efficient use of bandwidth
- More users in same spectrum

◆ 7. Mobility Support

- Works even at high speed (trains, cars)
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5. Real-world Applications

- Smart cities
 - Self-driving cars
 - IoT devices
 - Online gaming
 - Telemedicine
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6. Why 4G is Not Enough

Feature	4G	5G
Speed	Medium	Very High
Latency	High	Very Low
Devices	Limited	Massive
Reliability	Moderate	High
